ESR1 Structural Diversity and Redundancy Assessment

First month's report

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1. Project overview

We are studying how much the available ESR1 structures differ and how often similar shapes repeat. Our first step was to build a traceable dataset and compare corresponding protein coordinates.

This report covers **blocks 01–13: 13 of 22 Month 1 tasks (59.1%)**. Table 3 lists each reported task and its Python file. The master contains **487 ESR1 entries**; the baseline comparison uses **351 selected chains, one per applicable entry**.

Member	Reporting role	Work covered
Dana Koshkinbayeva	Data collection and quality	IDs, downloads and quality flags: 01, 02, 05.
Yerzhan Takauov	Domain and reference selection	Metadata, comparable domains and reference: 03, 04, 08.
Ayan Duisenbekov	Ligand and regional records	Identities, supported action labels and regional summaries: 06, 07, 13.
Ayaulym Bekenova	Structural comparison	Alignment, RMSD, groups and PCA: 09–12.

Table 1: Reporting responsibilities; the outputs are shared across the team.

2. Objectives

Collect and identify ESR1 structures, prepare comparable ligand-binding domains (LBDs), retain quality and ligand records, then establish baseline structural groups and regional summaries. Coverage here means implemented blocks with saved products; action classification remains incomplete.

3. Methods and tools

Inputs are PDB mmCIF structures, entry metadata and canonical ESR1 sequence correspondence. Entry, chain, model and ligand-instance IDs are kept separate. For example, `7NFB|B|1` identifies entry 7NFB, label chain B and model 1.

Parameter	Setting
Domain selection	Positions 305–550; ≥ 180 observed LBD and ≥ 60 core Cα atoms.
Combined quality flag	Resolution ≤ 2.5 Å; R-free ≤ 0.30; entry mean B ≤ 50 Å^2.
Reference selection	QC pass; $\geq 95\%$ LBD coverage; complete configured core and H12; best resolution, then coverage.
Alignment / pair distances	One saved Kabsch core fit per chain; ≥ 150 common LBD Cα positions per retained pair.
Structural groups / PCA	Average linkage, primary cut 1.5 Å; PCA of unscaled RMSD rows.
Regional B summaries	Within-chain population-SD normalisation; interval median; $\geq 80\%$ valid B coverage.

Table 2: Settings for the reported blocks. Domain eligibility and crystal quality are separate checks.

Tools: Biopython, NumPy, pandas, SciPy and Matplotlib for data processing and analysis; PyMOL for molecular visualization.

4. Progress and results

Reported tasks. The sequence below follows the checklist: collection and mapping first, then reference alignment, structural comparison and regional measurements.

Block	Checklist task	Main Python file	Saved product
01	Find structure IDs	`query_structures.py`	487-entry master
02	Download structures	`download_structures.py`	Download records, hashes
03	Extract metadata and molecular contents	`extract_metadata.py`	Entry/content inventory
04	Prepare comparable receptor domains	`extract_lbd.py`	Mapped chains, coverage
05	Add quality flags	`flag_quality.py`	QC labels and values
06	Build the ligand identity table	`build_ligand_table.py`	Identity/instance links
07	Assign and verify ligand action classes	`assign_action_classes.py`	Labels and support status
08	Select a reference structure	`choose_reference.py`	`7NFB|B|1`
09	Align receptor structures	`align_structures.py`	Saved fits, coordinates
10	Calculate structural distances	`calculate_rmsd.py`	Indexed RMSD matrix
11	Cluster receptor conformations	`cluster_structures.py`	Baseline groups
12	Show the conformational landscape	`structural_pca.py`	PCA coordinates/plots
13	Summarize regional B-factors and geometry	`regional_variation.py`	B/coordinate summaries

Table 3: Scope: blocks 01–13. Concise task titles are shown; `regional_bfactors.py` supports block 13. Later tasks are excluded.

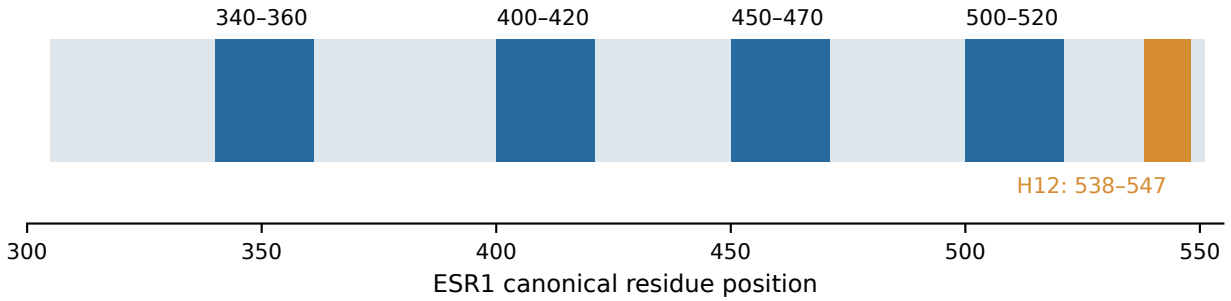


Figure 1: Configured fit windows contain up to 84 Cα positions. H12 is excluded from fitting so its position can be compared after the core alignment. This is a method sketch, not a full helix assignment.

Baseline measure	Result
ESR1 master / selected structural entries	487 / 351
Verified ESR1 chain-models / eligible copies	901 / 788
Selected entries passing combined QC	247 of 351 (70.4%)
ESR1 exact ligand identities / supported labels	400 / 7 (1.75%)
Structural matrix / distinct pair distances	351 × 351 / 61,425
Primary groups / entries in the two largest	9 / 336 of 351 (95.7%)

Table 4: Baseline results. Entry, chain-copy and ligand-identity counts use different denominators.

How to read the results. The following pages identify the relevant blocks before each result. Quality colours describe existing structures; they do not indicate a new clustering run. Structural groups organise coordinate similarity and do not assign biological activity.

4. Progress and results continued

01–04. Structure IDs, downloads, metadata and comparable domains. Of 487 master entries, 444 contain a verified ESR1 chain and 351 supply a selected comparable LBD. The matrix uses one selected chain per entry rather than all 788 eligible copies.

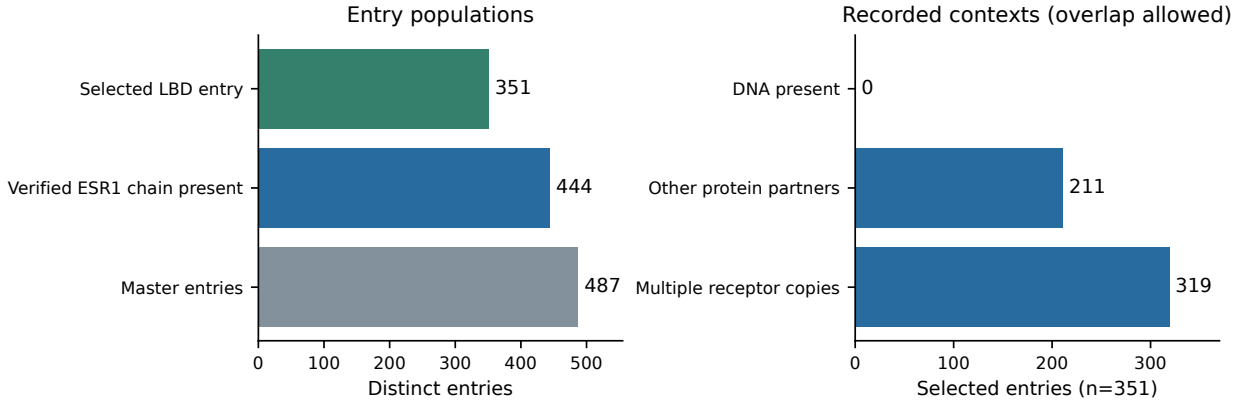


Figure 2: Entry populations and recorded contents. Counts on the left are nested; contexts on the right can overlap. Among selected entries, 319 have multiple receptor copies, 211 have other protein partners and none have DNA. Partner atoms are excluded from receptor RMSD.

05. Add quality flags. There are 247 QC passes and 104 failures among selected entries. Across the master, 369 pass, 117 fail and one is unknown. Of 136 entries outside the structural set, 122 pass QC: acceptable crystal quality alone does not provide a suitable LBD.

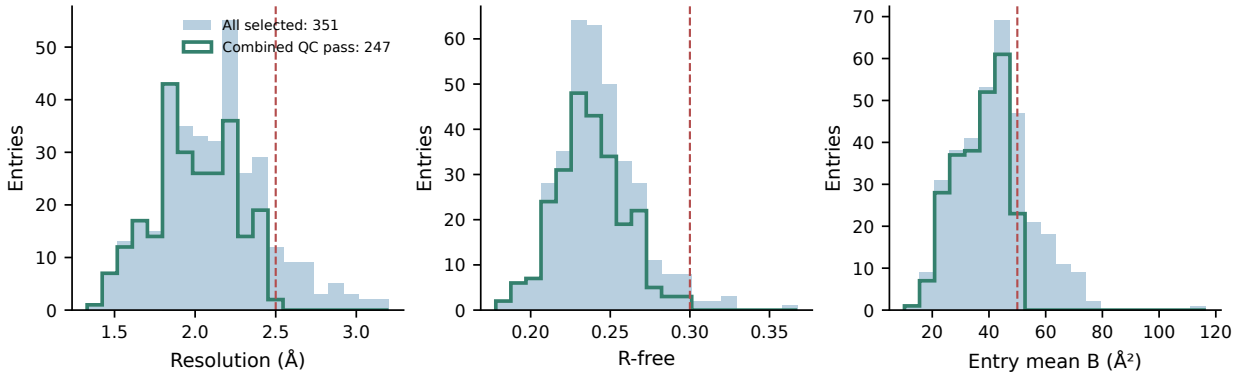


Figure 3: Quality distributions for 351 selected entries and their 247 combined-QC passes. Dashed lines show thresholds. A pass requires all three conditions; missing values are not automatic passes.

Practical outcome. Every master entry remains recorded, including fragments and entries excluded from the matrix. Missing residues are retained as missing rather than filled with invented coordinates.

4. Progress and results continued

06–07. Build the ligand identity table; assign and verify ligand action classes. The ESR1 table contains 400 exact identities. Seven have supported action labels: three agonists and four degraders. The other 393 remain unknown under the current evidence policy.

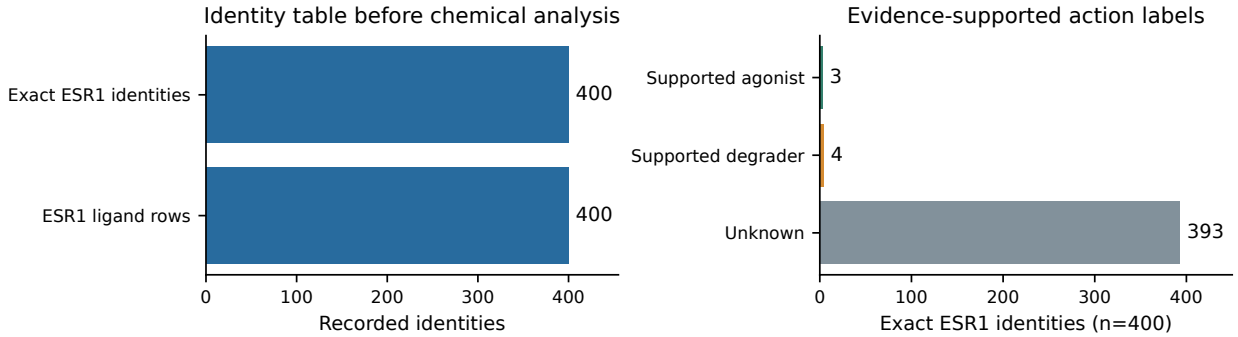


Figure 4: Identity and action-evidence coverage. Supported labels require validated evidence. Unknown means insufficient classification in these records, not absence of biological action.

08. Select a reference structure. Reference `7NFB|B|1` has resolution 1.33 Å, R-free 0.1781 and mean entry B 26.75 Å^2. It contains 244/246 expected LBD, 84/84 core and 10/10 configured H12 Cα positions. Its 28 annotated sequence differences remain a limitation.

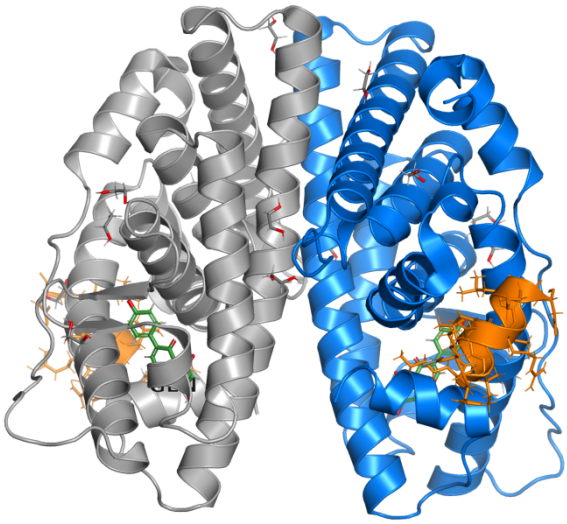


Figure 5: Reference-entry context: selected receptor blue, other polymers grey and short peptides orange. 7NFB contains two receptor copies and two partner chains. Green genistein carbons are shown across the asymmetric unit, including other copies; they are not an exact ligand-instance selection for chain B. Baseline image cropped for layout.

4. Progress and results continued

09. Align receptor structures. The saved Kabsch fit removes rigid translation and rotation using the configured core. Applying that transform to the domain preserves differences outside the fit, including H12.

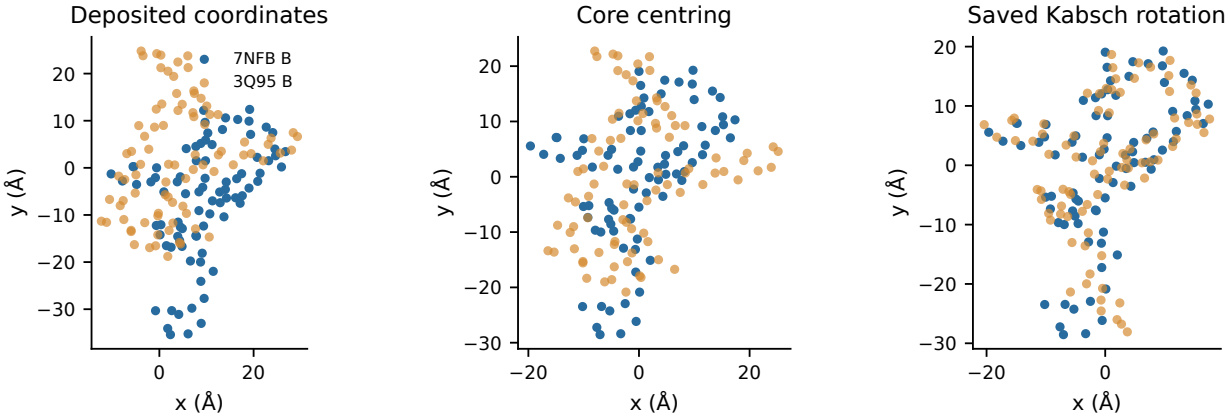


Figure 6: 3Q95 B aligned to 7NFB B using 84 shared core Cα atoms, projected onto two axes. Panels show deposited coordinates, core centring and the saved rotation. Rigid fitting changes the coordinate frame, not protein shape.

10. Calculate structural distances. The symmetric 351 × 351 matrix has a zero diagonal and 61,425 distinct pairs. Within/between-group median RMSDs are 0.800/3.820 Å. Because groups were derived from the same distances, this separation is descriptive rather than independent validation.

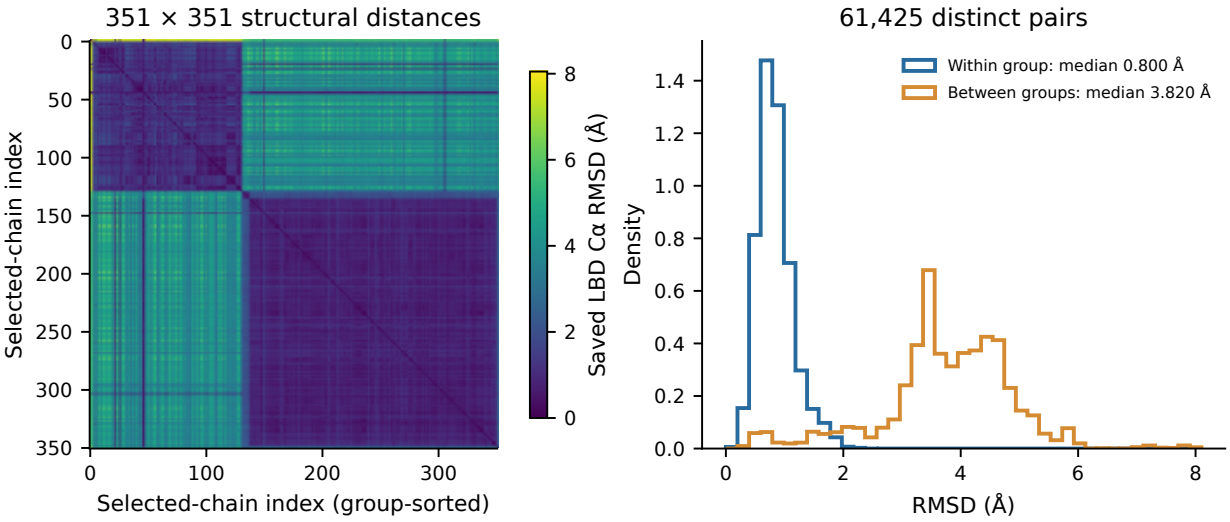


Figure 7: Saved LBD Cα RMSD after the core fit, with at least 150 common mapped positions per pair. Heatmap axes use the same group-sorted chain order; the histogram uses each distinct pair once. Pairs are not independent biological replicates.

Interpretation. Similar coordinates do not establish identical activity. Large distances identify structures to inspect alongside coverage, construct differences and experimental context.

4. Progress and results continued

11. Cluster receptor conformations. At the 1.5 Å cut there are nine groups. Groups 4 and 7 contain 124 and 212 entries: together 95.7% of the selected set.

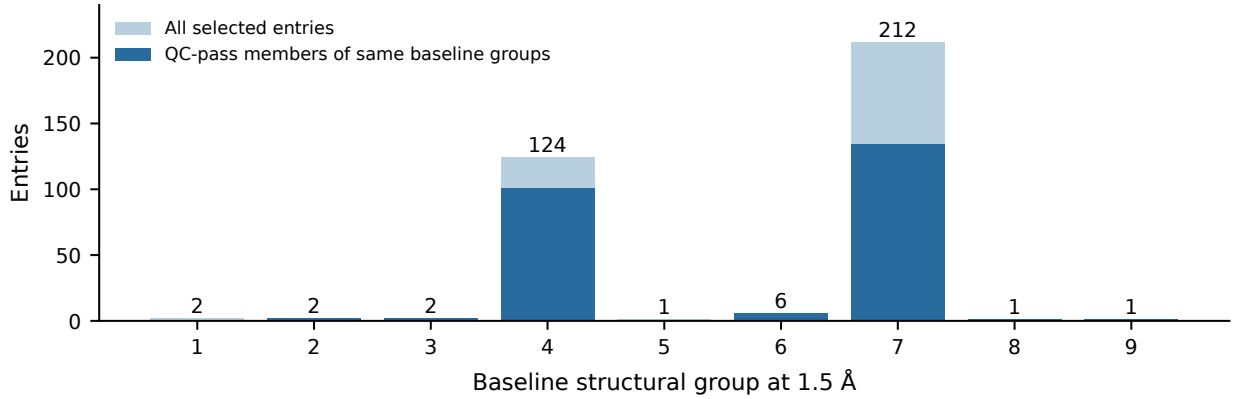


Figure 8: Baseline group counts; dark bars show QC-pass members of existing groups. No reclustering. Group IDs have no biological order.

12. Show the conformational landscape. PC1/PC2 explain 90.6%/7.5% of RMSD-row variance; both panels use the same coordinates.

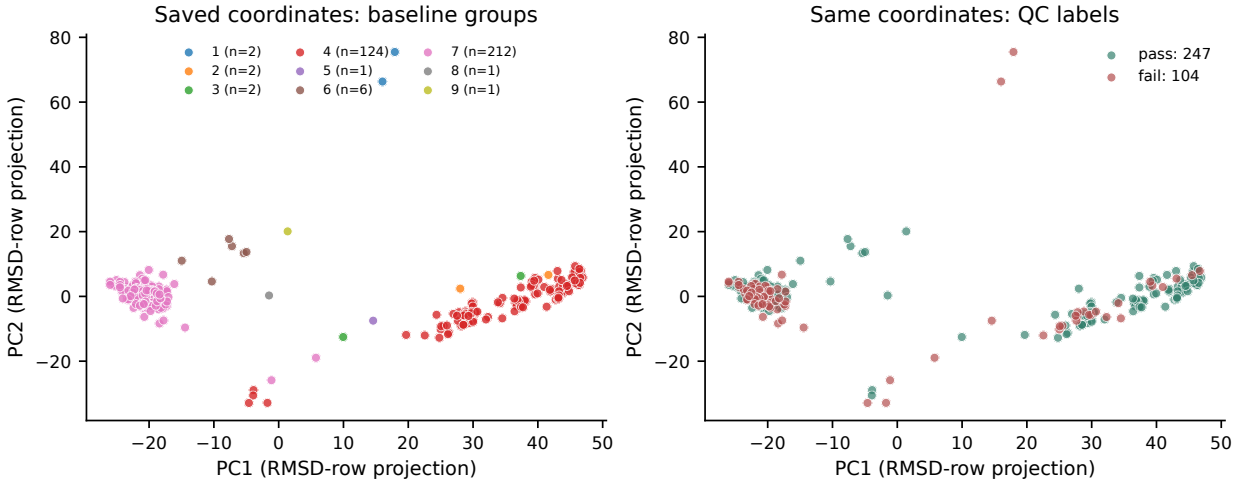


Figure 9: PCA coloured by group and QC. Axes project distance rows, not movement or free energy. Rare groups require inspection.

13. Summarize regional B-factors and geometry: relative B. Valid B values are normalised within each chain before comparing interval medians.

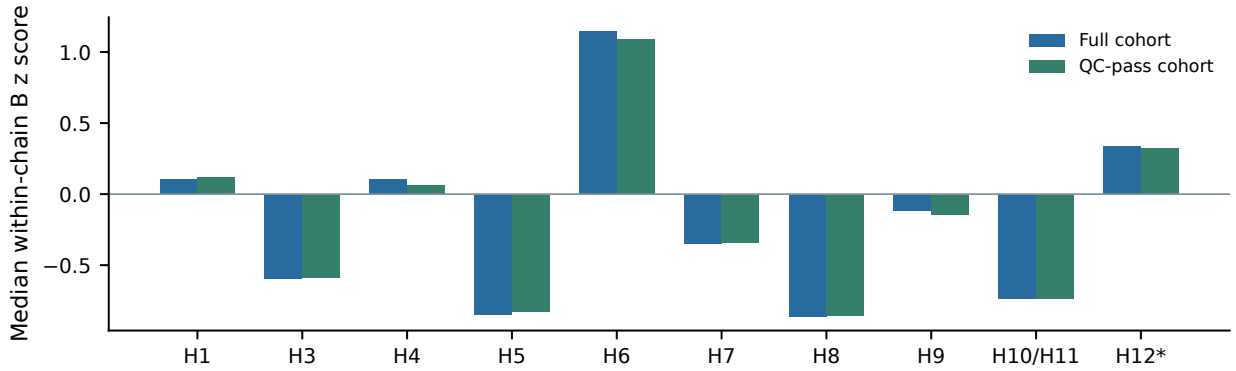


Figure 10: Regional B z scores with ≥ 80% valid B coverage. Boundaries are provisional: H12* 536–545; H2 unassigned; H10/H11 combined.

4. Progress and results continued

13. Summarize regional B-factors and geometry: coordinate variation. RMSF measures variation around the mean of aligned experimental structures, not motion along a simulation trajectory.

Interval	B chains	Full median z	QC median z	Variation (Å)
H6, 412–416	328	1.144	1.086	0.703
H12*, 536–545	344	0.339	0.325	6.500
H5, 371–395	351	−0.846	−0.830	0.423
H8, 441–456	351	−0.860	−0.857	0.334

Table 5: Regional results. B-chain counts use the full cohort and 80% valid-B rule. Variation is the root mean square of per-position ensemble RMSF, using available observations.

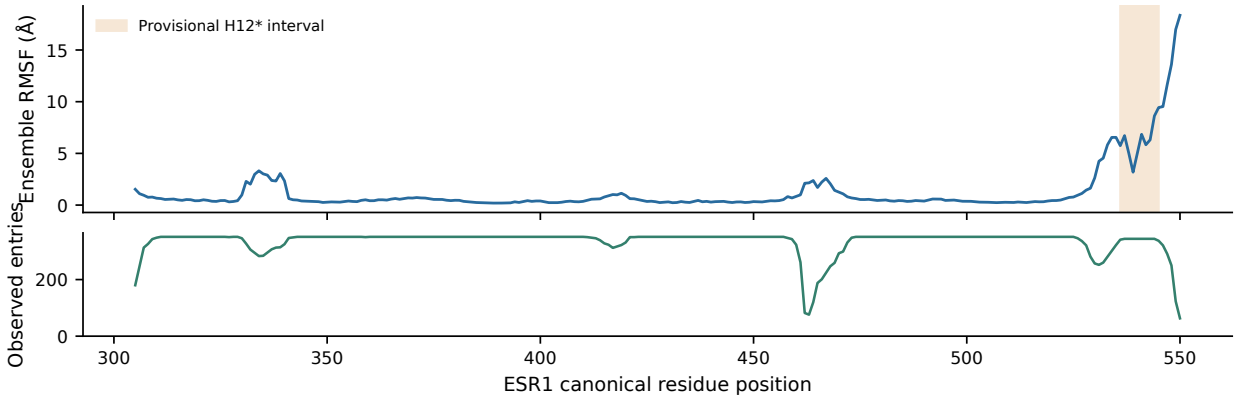


Figure 11: Per-position RMSF and observation counts after the saved core fit. Missing positions are excluded individually. H12* has large coordinate variation; coverage falls near the domain end.

H6 has the highest relative B, while H12* has the largest reported coordinate variation. These measurements have different meanings; neither alone establishes a biological mechanism.

5. Technical challenges

Multiple copies, chain identifiers and missing residues required explicit mapping. Action evidence remains incomplete. The reference has sequence differences; helix boundaries are provisional. B-factors and coordinate variation should not be interpreted as direct proof of motion.

6. Next steps

Review unresolved action labels, confirm regional boundaries and inspect rare groups against sequence, coverage and quality records. Additional checklist tasks remain outside this report's scope.

7. Documentation

Table 3 maps tasks to Python files. RMSD arrays are stored with ordered chain indices; metadata retain entry, chain and model identifiers. Regional tables include observation counts and coverage. The reported figures use these saved analysis outputs.

8. Additional notes

We practised mmCIF parsing, sequence mapping, rigid fitting, matrix indexing and plotting. Our main lesson was to separate what the code measures from the biological conclusions that still need evidence.